Sibabrata Biswal

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Experience

100x Engineers, Applied GenAI Cohort — Remote, Ongoing

Applied Generative AI and AI Engineering

- Built and experimented with **Generative AI applications** using **LLMs**, **RAG pipelines**, and **tool-calling frameworks**.
- Learned and implemented concepts in **diffusion models**, **AI automations**, **Model Context Protocol (MCP)**, and **GenAI application architecture**.
- Developed understanding of **AI agents** and **multi-agent workflows** for orchestrating autonomous AI systems.
- Worked with modern AI engineering stacks for building scalable and production-oriented AI applications.

Jawaharlal Nehru University(JNU), Research Associate — New Delhi,IN, Feb 2025 – Aug 2025

Prof. B. R. Panda, School of Biotechnology,JNU Website

AI-based Prediction of Metastasis in Tongue Cancer

- Fine-tuned **deep learning models** on **radiology data** for tumor classification and localization in head and neck cancer.
- Generated **Grad-CAM** and **attention-map** visualizations to interpret model predictions.

PreprintCommons – An Open Platform for Scientific Preprint Discovery

- Contributed in developing data collection pipeline using **open-source LLMs** to extract, clean, and structure metadata from preprints
- Designed and implemented a full-stack web application using **JavaScript, React, TailwindCSS, FastAPI** and **PostgreSQL** to support seamless preprint discovery and exploration.

Deepgene ,Center for Cellular And Molecular Platforms, Research Assistant — Bangalore, IN, Sept 2024 – Feb 2025

- Worked on the **PreprintCommons** and **Cancer Diagnosis** project under the mentorship of Prof. B. R. Panda; Project responsibilities and scope continued at **JNU**.

Jigyasu, DASCO Education Pvt Ltd, Academic Manager Website — Bhubaneswar, IN, May 2024 – Sept 2024

Job Description:

- Session Management
- Academic Operations
- Product/Content Development
- Client Technical Demonstrations
- Academic work in online digital branding

TiH-IoT Foundation, Indian Institute of Technology(IIT), Bombay, Project Fellow — Mumbai, IN, May 2023 – Jan 2024

Dr. Subhankar Mishra, School of Computer Sciences, NISER Website

Predictive Irrigation System Using Internet of Things(IoT) and Machine Learning

- Designed IoT-enabled plant monitoring system (*A. thaliana*) tracking soil moisture in a controlled soil nutrient,environmental parameters, developing **ML models** for irrigation prediction
- Contributed to literature review, experimental design, sensor selection, and hardware-software pipeline integration for predictive scheduled irrigation

Indian Council of Medical Research (ICMR), Government of India, Research Intern — Bhubaneswar, IN

Dr. Susheel Kumar Singh, ICMR-Regional Medical Research Centre Website — July 2022

AP205 VLP Vaccine Candidate Development Using *E. coli* Expression Systems

- Performed molecular cloning techniques including **plasmid design, transformation, and protein expression** in *E. coli*
- Conducted protein purification and characterization using **SDS-PAGE and Western blotting**

Publications

Book Chapter: — Nov 2024

Predictive irrigation: current practice and future prospects

Subhrajyoti Mishra , ***Sibabrata Biswal***, Anuleho Biswas, Abhijit Chakraborty, Subhankar Mishra

10.1016/B978-0-443-24139-0.00022-9

Conference proceeding: — Dec 2024

Iot4Irigation - Integrating IoT and Machine Learning for low-cost Sustainable Agriculture

10.1007/978-3-031-74701-4

Projects

Deep Learning-based Modeling for Predicting Drug-Kinase Bioactivity in Drug Repurposing [Master's Thesis]

Supervised by: **Dr. V Badireenath Konkimalla**, Drug discover lab, School of Biological Sciences, NISER Website

- Developed **GNN-based drug-kinase interaction predictor** using ChEMBL data, with heterogeneous graphs combining **RDKit descriptors** and **ProtBert embeddings** (MAE:1.02, RMSE:1.28)
- Implemented end-to-end pipeline in **PyTorch** with **Deep Graph Library (DGL)**, featuring an encoder-decoder architecture with stratified graph splitting for robust **drug repurposing** predictions

Integrative Analysis of miRNA-mRNA Network Reveals Potential Biomarkers for ALK-Rearrangement Associated Lung Adenocarcinoma

Supervised by: **Dr. V Badireenath Konkimalla**, Drug discover lab, School of Biological Sciences, NISER Website

- Identified 15 **miRNA-mRNA biomarker candidates** for ALK+ lung adenocarcinoma via **multi-omics integration** (limma / WGCNA / STRING),revealing **dysregulated pathways** (E2F targets, G2M checkpoint) through **functional enrichment** (GO/KEGG)
- Developed **R/Bioconductor pipeline** with **Cytoscape visualization**, achieving **0.82 AUC** in ROC analysis for top biomarkers demonstrating clinical potential for **personalized cancer therapy**

Deep Learning-Driven Discovery of Phytochemicals Targeting BRAF Mutations for Precision Cancer Therapy

Supervised by: **Dr. V Badireenath Konkimalla**, Drug discover lab, School of Biological Sciences, NISER Website

- Developed a **Chemprop-based deep learning model** (AUC: 0.967) to screen **2,846 phytochemicals** from PubChem (via **RDKit-processed SMILES**), identifying high-potential BRAF inhibitors.

Education

Integrated Master of Sciences | Life Sciences — 2019 - 2024

National Institute of Science Education and Research (NISER), Bhubaneswar

- CGPA: 7.55/10.0

Intermediate (+2) | Science — 2017

SSVM, Neelakanthanagar, Brahmapur

- Percentage: 73%

Matriculation 2015
SSVM, Gosainnuagao, Brahmapur, Odisha

- Percentage: 85.67%

Conferences

International Conference on Intelligent Computing and Big Data Analytics (ICICBDA) June 2024
Mumbai, IN
National Conference for Undergraduate Research in Biosciences Dec 2024
Tirupati, IN
EAGxIndia (Effective Altruism Global) Oct 2024
Bangalore, IN

Technical Skills

Programming:	C, Python, JavaScript, R, SQL, Bash
Frameworks:	React, PyTorch, Scikit-learn, CSS, Tailwind
Bioinformatics:	PyMOL, RDKit, GROMACS, AutoDock, Vina
Data Analysis:	RNA-Seq, WGBS, MNase-Seq, WGCNA, Cytoscape
Tools:	LaTeX, Notion, Photoshop, Illustrator, Blender
OS:	Windows, Linux

Training & Certifications

Biosecurity Fundamentals 2023
BlueDot Impact

- Covered global catastrophic biological risks and mitigation strategies

CS50x: Introduction to Computer Science 2023
Harvard University (edX)

Achievements

DISHA Scholar 2019-2024
Department of Atomic Energy, Govt. of India
CHANAKYA Fellowship 2023-2024
National Mission on Interdisciplinary Cyber Physical Systems (NM-ICPS) , Govt. of India
GATE-Life Science Qualifier 2025
Graduate Aptitude Test in Engineering
NEST Qualifier 2019
National Entrance Screening Test